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High-Dimensional Knockoffs Inference for Time Series Data

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ABSTRACT

We make some initial attempt to establish the theoretical and methodological foundation for the model-X knockoffs inference for time series data. We suggest the method of time series knockoffs inference (TSKI) by exploiting the ideas of subsampling and e-values to address the difficulty caused by the serial dependence. We also generalize the robust knockoffs inference in Barber, Candès, and Samworth to the time series setting to relax the assumption of known covariate distribution required by model-X knockoffs, since such an assumption is overly stringent for time series data. We establish sufficient conditions under which TSKI achieves the asymptotic false discovery rate (FDR) control. Our technical analysis reveals the effects of serial dependence and unknown covariate distribution on the FDR control. We conduct a power analysis of TSKI using the Lasso coefficient difference knockoff statistic under the generalized linear time series models. The finite-sample performance of TSKI is illustrated with several simulation examples and an economic inflation study. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

Identifying key economic factors among a large number of potential variables (e.g., numerous types of consumer price indices, unemployment rates, and housing prices) that can influence inflation is a long-standing research pursuit (King, Stock, and Watson 1995; Stock and Watson 1999; Crump et al. 2022) that remains crucial due to inflation's significance. However, statistical inference for economic time series such as inflation is challenging due to the serial dependence, large number of potentially important covariates (including time series covariates, their lags, and non-time series covariates), regime shifts (Hamilton 1989; Tong and Lim 1980), and possible non-linear relationships.

Let us exemplify these challenges with Figure 1, which depicts the monthly inflation rate (hereafter referred to as inflation) time series of the U.S. economy from May 2013 to January 2023, sourced from the FRED-MD database (McCracken and Ng 2016) and the U.S. Bureau of Labor Statistics. In addition to the inflation series, the FRED-MD database includes 126 other time series variables that can be used to predict the inflation for the following month. Here, inflation is calculated as the first-order difference of the Consumer Price Index (CPI) (McCracken and Ng 2016) divided by the CPI of the previous month; see Section 5 for details. Figure 1 shows that during the period from 2020 to late 2022, inflation displayed an upward mean drift due to the U.S. economy's post-COVID-19 recovery and the impacts of

the Russia–Ukraine conflict, along with some possible stationary phase-changing behaviors (Hamilton 1989; Tong and Lim 1980). To address concerns on potential nonstationarity over the entire time span caused by, say, the mean drift, the rolling-window method is commonly employed in time series analysis. The intuition behind this method is that the time series, including both the response and predictors, are more likely to be stationary within a small time window.¹ However, the use of rolling windows further complicates the problem in that the sample size in each window is usually small; for example, there are only 60 data points if sampled monthly over a five-year period. The presence of serial dependence, small sample size, high-dimensional covariates, and possibly nonlinear relationships makes statistical inference for time series regression highly challenging. Variable selection has been a popular solution to address such challenges, under the assumption that only a small subset of covariates contribute to the response of interest. Correctly selecting these important covariates can help simplify the model and improve interpretability and prediction accuracy. Our goal in this article is to develop a reliable variable selection approach that accounts for these unique challenges in time series regression.

Popularly used measures for evaluating the performance of high-dimensional variable selection include the false discovery rate (FDR) (Benjamini and Hochberg 1995), model selection consistency (Zhang 2011; Medeiros and Mendes 2016; Ing 2020), and feature importance ranking (Breiman 2001; Medeiros et al. 2021). Our article focuses on controlling the FDR, whose

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¹ We also obtain numerical evidence supporting the stationarity of the inflation within each window through standard unit root tests; details are provided in Section A.5.

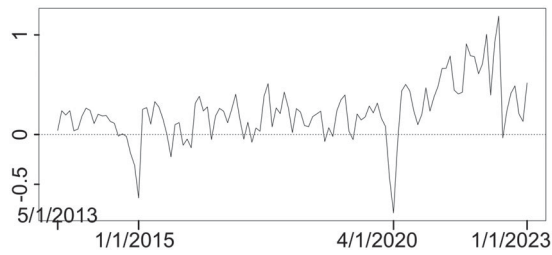


Figure 1. The U.S. inflation from May 2013 to January 2023.

formal definition is in (1). Most existing works address the FDR control by building procedures based on the p -values constructed for assessing the importance of individual variables; see, for example, the seminal works of Benjamini and Hochberg (1995); Benjamini and Yekutieli (2001). Yet, the high dimensionality in covariates and the possibly complicated nonlinear model structure make many conventional ways of p -value calculations inapplicable or even completely fail (Fan, Demirkaya, and Lv 2019). To overcome such difficulties, the framework of knockoffs inference was proposed in Barber and Candès (2015); Candès et al. (2018) to achieve the goal of exact FDR control in variable selection in finite samples, completely bypassing the use of the conventional p -values in high-dimensional regression models. It allows for an arbitrary dependence structure of the response on covariates and an arbitrary dimensionality of covariates at the cost of assuming the known joint covariate distribution. See Section 2.2 for a brief review of the model-X knockoffs framework.

Two critical assumptions in model-X knockoffs (Candès et al. 2018) are: (a) the observations across time are iid and (b) the joint distribution of covariate vector is known for generating the knockoff variables. Both assumptions are unreasonably strong for time series data. Time series data exhibit serial dependence, and in some applications where covariates are lagged response variables with a stationarity assumption, assuming a known covariate distribution directly reveals the set of important variables, thus invalidating the problem of variable selection. We will address these challenges by relaxing the two aforementioned assumptions.

To relax the first assumption, we adopt the popular idea of subsampling. To relax the second assumption, we generalize the robust knockoffs inference in Barber, Candès, and Samworth (2020) to the time series setting, where the robust knockoffs inference (Barber, Candès, and Samworth 2020) is a recent innovation that allows for approximate covariate distribution (as opposed to known covariate distribution) developed for iid data. We name the knockoff variables generated using approximate covariate distribution as *approximate knockoffs* to ease the presentation. Our analyses reveal that with approximate knockoffs generated in a rowwise fashion ignoring the serial dependence, the FDR inflation has an upper bound depending on the Kullback–Leibler (KL) divergence between the distributions of data matrices corresponding to the approximate and exact model-X knockoff variables. Our theoretical results are comparable to Theorem 1 of Barber, Candès, and Samworth (2020), with the difference that we do not need the iid observations assumption. Our theory shows that there is generally no guarantee that such KL divergences asymptotically vanish in the

existence of serial dependence, suggesting that the corresponding FDR could be uncontrolled. We also show that subsampling (Yu 1994) can successfully address such difficulty and warrant asymptotically vanishing KL divergence if the subsampling rate is appropriately chosen, provided that the time series are β -mixing. We then apply the robust knockoffs inference to each of the subsampled data, resulting in multiple sets of selected variables. We aggregate these sets via the e-value method (Wang and Ramdas 2022; Ren and Barber 2022). The complete framework is presented in Section 2 and named as the *time series knockoffs inference* (TSKI). We provide a rigorous characterization of how the serial dependence and the accuracy of the approximate knockoffs affect the FDR control and prove that the TSKI procedure can achieve asymptotic FDR control when the subsampling is done appropriately and the approximate knockoffs are accurate enough.

It is well-known that FDR and power are two sides of the same coin. We then turn to the power analysis of TSKI in Section 3. Assuming the generalized linear time series models and some regularity conditions, we show that for TSKI with subsampling and e-value aggregation, the set of selected variables is either empty or enjoys the sure screening property with asymptotic probability one. These results are formally summarized in Theorem 2 and discussed after it.

We test the empirical performance of TSKI on both simulated and real data. Our simulation results in Section 4 demonstrate that the TSKI with subsampling controls the FDR when the data are generated from some popular β -mixing processes. The selection power is generally satisfactory with a sufficient sample size. Furthermore, in Section 5 we apply the TSKI to study the temporal relations between inflation and other macroeconomic time series from the U.S. economy over the past ten years.

1.1. Related Work

Established methods such as the BH and BY (Benjamini and Hochberg 1995; Benjamini and Yekutieli 2001) are commonly used in biology applications with non-time series data. Some new developments (e.g., Ramdas et al. 2019; Bogdan et al. 2015) either assume independent test statistics, require the availability of p -values, or rely on specific model structures, making them unsuitable for time series applications due to incompatible assumptions.

Among these existing methods, BY achieves the FDR control in variable selection based on a set of valid p -values with no requirements on the dependence structure among p -values. Similarly, e-BH (Wang and Ramdas 2022) uses e-values without requirements on their dependence structure. While these methods offer the potential for valid FDR control for time series data, obtaining valid p -values or e-values remains an unresolved challenge in many applications.

Regularized regression (Medeiros and Mendes 2016) and information criterion-based model selection (Ing 2020) are also popularly used for selecting important variables in time series regressions. Yet, they often assume some specific model structure (e.g., the linear model) to prove variable/model selection consistency, posing uncertainty about their performance when deployed on real data with nonlinear dependency. In addition,

they are not specifically designed for controlling the variable selection error rate, and hence may not be suitable for certain applications when the error rate control is a concern.

1.2. Notation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{R}$ be the underlying probability space and the Borel σ -algebra on the real line $\mathbb{R}$, respectively. We use the boldface for random vectors and matrices, the tilde for the knockoff variables, and the vector notation $\tilde{x}$ to denote vectors in the Euclidean space. For random vector $\mathbf{x}$, define $\mathbf{x}_{-j}$ as the subvector by removing the j th coordinate. We use parentheses for matrix concatenation. For any real sequences $\{a_n\}$ and $\{b_n\}$, $a_n = O(b_n)$ means $\limsup_{n \rightarrow \infty} \left| \frac{a_n}{b_n} \right| < \infty$, and $a_n = o(b_n)$ means $\limsup_{n \rightarrow \infty} \left| \frac{a_n}{b_n} \right| = 0$. We use $\#S$ to denote the cardinality of a given set S . Moreover, a transition kernel is defined as a map $p : (\mathbb{R}^{k_1}, \mathcal{R}^{k_2}) \rightarrow [0, 1]$ for some positive integers k_1 and k_2 satisfying that (i) for each $\mathcal{D} \in \mathcal{R}^{k_2}$, $p(\cdot, \mathcal{D})$ is a measurable function and (ii) for each $x \in \mathbb{R}^{k_1}$, $p(x, \cdot)$ is a probability measure.

2. Robust Time Series Knockoffs Inference with TSKI

Given an observed stationary time series $\{Y_t, \mathbf{x}_t\}_{t=1}^n$ with $Y_t \in \mathbb{R}$ a scalar response and $\mathbf{x}_t \in \mathbb{R}^p$ a high-dimensional covariate vector, we are interested in accurately selecting relevant covariates (i.e., non-null features) in $\mathbf{x}_t$, where the definition of the null feature is given below.

Definition 1 (Null feature). Consider the response Y and the covariate vector $\mathbf{x} = (X_1, \dots, X_p)^T$. Covariate X_j with $j \in \{1, \dots, p\}$ is said to be null with respect to response Y if and only if $X_j \perp\!\!\!\perp Y | (X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_p)^T$.

We denote the set of null features according to Definition 1 as $\mathcal{H}_0 \subset \{1, \dots, p\}$ for each t , where $\mathcal{H}_0$ is independent of t because $(Y_t, \mathbf{x}_t)$'s have the same distribution due to the stationarity assumption. Our goal is to estimate the important variable set $\mathcal{H}_1 = \{1, \dots, p\} \setminus \mathcal{H}_0$ using data $\{Y_t, \mathbf{x}_t\}_{t=1}^n$. For an estimated set $\hat{S} \subset \{1, \dots, p\}$ returned by some algorithm, we measure the accuracy by evaluating the False Discovery Rate (FDR) defined as

$$\text{FDR} := \mathbb{E} \left(\frac{\#(\hat{S} \cap \mathcal{H}_0)}{(\#(\hat{S}) \vee 1)} \right). \quad (1)$$

In this article, we focus on time series data with serial dependency and high dimensionality where covariate dimensionality p can be much larger than sample size n . We do not assume any specific dependence structure of Y_t on $\mathbf{x}_t$ other than the one in Definition 1 and that Y_t is $\mathbf{x}_{t+1}$ -measurable (see Corollary 2) for our FDR analysis. As a result, our proposed method can accommodate an unknown relationship between Y_t and $\mathbf{x}_t$ including both linear and nonlinear ones. Our method builds on the recent work of model-X knockoffs (Candès et al. 2018) and its robust extension (Barber, Candès, and Samworth 2020) proposed for the non-time-series data, where we briefly review the former in the next section to set the stage.

2.1. A Brief Review of the Model-X Knockoffs Framework

Let $\mathbf{y} = (Y_1, \dots, Y_n)^T \in \mathbb{R}^n$ and $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^T \in \mathbb{R}^{n \times p}$ be the response vector and design matrix collecting the n observations. The model-X knockoffs (Candès et al. 2018) was proposed for the setting where the rows of the augmented matrix $(\mathbf{y}, \mathbf{X}) \in \mathbb{R}^{n \times (p+1)}$ are iid random vectors with known distribution for $\mathbf{x}_1$. The knockoffs inference aims at estimating $\mathcal{H}_1$ while keeping the FDR (1) under control. To this end, it constructs an $n \times p$ matrix $\tilde{\mathbf{X}}$ in a rowwise fashion independently using the known joint distribution of $\mathbf{x}_1$ such that

$$\tilde{\mathbf{X}} \perp\!\!\!\perp \mathbf{y} | \mathbf{X} \text{ and } (\mathbf{X}, \tilde{\mathbf{X}})_{\text{swap}(S)} \stackrel{d}{=} (\mathbf{X}, \tilde{\mathbf{X}}) \quad (2)$$

for each $S \subset \{1, \dots, p\}$, where $\text{swap}(S)$ denotes the swapping operation meaning that for each $j \in S$, columns j and $j + p$ are swapped, and $\stackrel{d}{=}$ stands for equal in distribution. At a high level, the model-X knockoffs create a “fake” covariate matrix $\tilde{\mathbf{X}}$ which perfectly mimics the “behavior” of the original covariate matrix. By using these fake covariates as controls, the importance of original covariates can be inferred.

As an example, when the covariate distribution for $\mathbf{x}_t$ is known to be $N(\mathbf{0}, \Sigma)$ with Σ the $p \times p$ covariance matrix, a valid way to construct the corresponding ideal knockoff vector $\tilde{\mathbf{x}}_t$ is to sample from the conditional multivariate Gaussian distribution

$$\tilde{\mathbf{x}}_t | \mathbf{x}_t \stackrel{d}{\sim} N(\mathbf{x}_t - \text{diag}(\tilde{s}) \Sigma^{-1} \mathbf{x}_t, 2 \text{diag}(\tilde{s}) - \text{diag}(\tilde{s}) \Sigma^{-1} \text{diag}(\tilde{s})), \quad (3)$$

where $\text{diag}(\tilde{s})$ is a diagonal matrix of tuning parameters with positive diagonal entries. Larger components of $\tilde{s}$ imply that the resulting knockoff variables are more independent of the original variables, thereby providing higher power in distinguishing them. For the general covariate distribution, a conditional distribution for generating knockoff variables can also be constructed following the same high-level idea above. Further details about knockoff variable sampling procedure for general distributions can be found in Candès et al. (2018) and Fan et al. (2020).

With the knockoff variable matrix $\tilde{\mathbf{X}}$, the knockoff statistics W_j 's, measuring the importance of original covariates, are constructed such that the sign-flip property (Barber and Candès 2015; Candès et al. 2018) is satisfied: for each $S \subset \{1, \dots, p\}$ and each $1 \leq j \leq p$,

$$W_j(\mathbf{y}, [\mathbf{X}, \tilde{\mathbf{X}}]_{\text{swap}(S)}) = \begin{cases} -W_j(\mathbf{y}, \mathbf{X}, \tilde{\mathbf{X}}) & \text{if } j \in S, \\ W_j(\mathbf{y}, \mathbf{X}, \tilde{\mathbf{X}}) & \text{otherwise.} \end{cases} \quad (4)$$

As variable importance measures, high-quality knockoff statistics W_j 's should have the desired properties that 1) W_j 's have large positive values for $j \in \mathcal{H}_1$ and 2) for null features $j \in \mathcal{H}_0$, W_j 's have small magnitude and are symmetric around zero. See Candès et al. (2018) for the formal characterization. Examples 1–2 in the next subsection are two important instances of the knockoff statistics that satisfy the sign-flip property.

Model-X knockoffs (Candès et al. 2018) estimates $\mathcal{H}_1$ as $\hat{S} = \{1 \leq j \leq p : W_j \geq T\}$ with the so-called knockoff threshold $T = \min \left\{ t > 0 : \frac{1 + \#\{j: W_j \leq -t\}}{\#\{j: W_j \geq t\} \vee 1} \leq \tau \right\}$, where $\tau \in (0, 1)$ is some pre-specified target level for FDR control. Candès et al. (2018)

Algorithm 1: Robust time series knockoffs inference (TSKI) via e-values

- 1 Let $0 < \tau_1 < 1$ be a constant and $0 < \tau^* < 1$ the target FDR level.
- 2 For each $k \in \{1, \dots, q+1\}$, calculate the knockoff statistics $W_1^k, \dots, W_p^k$ satisfying (4) using sample $\{\mathbf{x}_t, \tilde{\mathbf{x}}_t, Y_t\}_{t \in H_k}$.
- 3 Calculate the e-value statistics $e_j = (q+1)^{-1} \sum_{k=1}^{q+1} e_j^k$, where²

$$e_j^k = \frac{p \times \mathbf{1}_{\{W_j^k \geq T^k\}}}{1 + \sum_{s=1}^p \mathbf{1}_{\{W_s^k \leq -T^k\}}}, \quad T^k = \min \left\{ t \in \mathcal{W}_+^k : \frac{1 + \#\{j : W_j^k \leq -t\}}{\#\{j : W_j^k \geq t\} \vee 1} \leq \tau_1 \right\}, \quad (5)$$

and $\mathcal{W}_+^k = \{j : W_j^k > 0\}$ for each $k \in \{1, \dots, q+1\}$. Here, $\mathbf{1}_{\{\cdot\}}$ is the indicator function.

- 4 Let $\widehat{S} = \{j : e_j \geq p(\tau^* \times \widehat{k})^{-1}\}$ with $\widehat{k} = \max\{k : e_{(k)} \geq p(\tau^* \times k)^{-1}\}$, where $e_{(j)}$'s are the ordered statistics of e_j 's such that $e_{(1)} \geq \dots \geq e_{(p)}$.

showed that the model-X knockoffs framework achieves FDR control in finite samples with arbitrary dimensionality of $\mathbf{x}_t$ and arbitrary (unknown) dependence structure of Y_t on $\mathbf{x}_t$.

As discussed in the Introduction, the iid row assumption for $(\mathbf{y}, \mathbf{X})$ and the known distribution assumption for $\mathbf{x}_t$ are too stringent for time series data. We will develop a new framework for time series knockoffs inference, which relaxes these assumptions. It is important to note that the remaining part of the article does *not* assume iid rows in data matrix $(\mathbf{y}, \mathbf{X})$.

2.2. Outline of the TSKI Framework

In this section, we provide an outline of the TSKI framework. Some technical details will be presented in the next subsection. TSKI has three key ingredients: subsampling, robust knockoffs inference on each subsample, and e-value aggregation of the selected sets from different subsamples. Algorithm 1 provides a detailed implementation of TSKI. Our presentation may not strictly adhere to the order of the three ingredients stated above.

To relax the known covariate distribution assumption, we adopt the robust knockoffs framework in Barber, Candès, and Samworth (2020) and introduce the following Definition 2. In what follows, the knockoff generator, conditional distribution, and transition kernel are examples of regular conditional probability (r.c.p.) in probability theory.

Definition 2 (Knockoff generator). $\kappa : \mathbb{R}^p \times \mathcal{R}^p \mapsto \mathbb{R}^p$ is said to be a knockoff generator if 1) $\kappa(\tilde{\mathbf{z}}, \cdot)$ is a probability measure for each $\tilde{\mathbf{z}} \in \mathbb{R}^p$ and 2) $\kappa(\cdot, \mathcal{A})$ is a measurable function for each $\mathcal{A} \in \mathcal{R}^p$, where $\mathcal{R}$ is the smallest σ -algebra of $\mathbb{R}$ that contains all open sets.

For each observed covariate vector $\mathbf{x}_t$ with $t \in [n] := \{1, \dots, n\}$, we generate its knockoff vector $\tilde{\mathbf{x}}_t$ from $\kappa(\mathbf{x}_t, \cdot)$. When $\mathbf{x}_t$ has known distribution $N(\mathbf{0}, \Sigma)$ with known parameters Σ , the conditional Gaussian distribution in (3) is an instance of the knockoff generator which yields *ideal* knockoff variables satisfying (2). If Σ in (3) is unknown and replaced with an estimated version (see (8) for details), the resulting knockoff generator produces only *approximate* knockoff variables that violate the exchangeability condition in (2). Additional examples of the knockoff generator can be found in (Candès et al. 2018; Fan et al. 2020). To achieve asymptotic FDR control using knockoff

variables generated from $\kappa(\mathbf{x}, \cdot)$, we need some additional conditions on κ which will be presented in Condition 3 in the next section. Note that because of the rowwise generation of knockoff variables, although $\{\mathbf{x}_t\}_{t=1}^n$ have serial dependence across t , $\{\tilde{\mathbf{x}}_t\}_{t=1}^n$ are independent across t conditional on $\{\mathbf{x}_t\}_{t=1}^n$; this violates the second property in (2) and thus the FDR control result in Candès et al. (2018) or Barber, Candès, and Samworth (2020) cannot be applied directly. From now on, we work with the inference sample $\{Y_t, \mathbf{x}_t, \tilde{\mathbf{x}}_t\}_{t=1}^n$.

To overcome such difficulty, we consider subsamples each with index set $H_k = \{k + s(q+1) : s = 0, 1, \dots, \lfloor \frac{n-k}{q+1} \rfloor\}$ for $k \in \{1, \dots, q+1\}$ with some integer $q \geq 0$. To simplify the technical presentation, let $\{V_t, \mathbf{u}_t, \tilde{\mathbf{u}}_t\}_{t=1}^N$ be a generic subsample that can be any of the $q+1$ subsamples. Denote by $\mathbf{v} := (V_1, \dots, V_N)^T$, $\mathbf{U} := (\mathbf{u}_1, \dots, \mathbf{u}_N)^T$, and $\tilde{\mathbf{U}} := (\tilde{\mathbf{u}}_1, \dots, \tilde{\mathbf{u}}_N)^T$. As in the robust knockoffs inference (Barber, Candès, and Samworth 2020), we construct knockoff statistics $W_j(\mathbf{v}, \mathbf{U}, \tilde{\mathbf{U}})$'s based on $(\mathbf{v}, \mathbf{U}, \tilde{\mathbf{U}})$ and select the set of important variables using these knockoff statistics following the identical procedure as reviewed in the last subsection. Thus, we end up with $q+1$ sets of selected variables $\{j : W_j^k \geq T^k\}$ with each corresponding to a subsample k . Here, W_j^k 's and T^k are the correspondingly constructed knockoff statistics and the knockoff threshold as specified in (5), respectively.

Below are two examples of the knockoff statistics. The random forests model in Example 2 can be replaced with other learning models such as the deep learning model.

Example 1 (Lasso coefficient difference (LCD)). For a given sample $(\mathbf{v}, \mathbf{U}, \tilde{\mathbf{U}})$ and a tuning parameter $\lambda \geq 0$, we define $W_j = W_j(\mathbf{v}, \mathbf{U}, \tilde{\mathbf{U}}) = |\hat{\beta}_j| - |\hat{\beta}_{j+p}|$, where $(\hat{\beta}_1, \dots, \hat{\beta}_{2p})^T$ is given by either the Lasso estimate $\arg \min_{\boldsymbol{\beta} \in \mathbb{R}^{2p}} \left\{ n^{-1} \sum_{t=1}^n (V_t - (\mathbf{u}_t^T, \tilde{\mathbf{u}}_t^T) \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{2p} |\beta_j| \right\}$ with $\boldsymbol{\beta} = (\beta_1, \dots, \beta_{2p})^T$, or the generalized linear model (GLM) Lasso estimate defined in (12).

Example 2 (Random forests mean decrease accuracy (MDA)). For a given sample $(\mathbf{v}, \mathbf{U}, \tilde{\mathbf{U}})$, we define $W_j = W_j(\mathbf{v}, \mathbf{U}, \tilde{\mathbf{U}}) = N^{-1} \sum_{t=1}^N \{ [V_t - \hat{m}(\mathbf{u}_t^{(j)}, \tilde{\mathbf{u}}_t^{(j)})]^2 - [V_t - \hat{m}(\mathbf{u}_t, \tilde{\mathbf{u}}_t^{(j)})]^2 \}$ for each $j \in \{1, \dots, p\}$, where $(\mathbf{u}_t^{(j)}, \tilde{\mathbf{u}}_t^{(j)}) = [\mathbf{u}_t, \tilde{\mathbf{u}}_t]_{\text{swap}(\{j\})}$ and $\hat{m} : \mathbb{R}^{2p} \mapsto \mathbb{R}$ is the random forests regression function trained by regressing V_t 's on $(\mathbf{u}_t, \tilde{\mathbf{u}}_t)$'s.

The LCD uses the linear model as a working model, while the MDA does not assume any explicit model structure. When the underlying true model is nonlinear, the LCD is based on the misspecified model, whereas the MDA is free of such issues. Yet, MDA demands a large sample size, which is a common drawback for all nonparametric regression models. Our simulation study will provide additional insights into the performance of LCD and MDA in various model settings.

When $q > 0$, subsampling yields more than one set of selected variables. Naively taking the intersection or union over these sets would not guarantee FDR control. The TSKI uses the e-BH procedure (Wang and Ramdas 2022) to overcome such a difficulty; see Step 3 in Algorithm 1 for the calculation of e-values and how it forms the final set of selected variables $\hat{S}$. The e-BH procedure works similarly to the BH procedure (Benjamini and Hochberg 1995) with the difference that e-values (Shafer 2021) are used in place of the p -values. Given a null hypothesis, we call a nonnegative random variable E an “e-value” if $\mathbb{E}[E] \leq 1$ under the null. To test a hypothesis at significance level α , we can reject the null hypothesis when $E \geq 1/\alpha$, noting that $\mathbb{P}(E \geq 1/\alpha) \leq \alpha \mathbb{E}[E] \leq \alpha$ under the null. One appealing property is that the average of multiple e-values is still a valid e-value regardless of their dependence structure. This motivates us to use it to aggregate results from different subsamples. We acknowledge that the e-value idea has been used in the literature for aggregating knockoffs inference results (Wang and Ramdas 2022; Ren and Barber 2022). We note that tuning parameter τ_1 used for individual subsample in Step 3 should be set smaller than the overall target level τ^* in Step 4 in order to achieve high selection power; this is formally stated in Theorem 2 in Section 3.

2.3. FDR Control by TSKI

We need some technical conditions to bound the FDR of TSKI.

Condition 1. The density function of $(X, \tilde{X}, Y)$ exists and $(Y_t, \mathbf{x}_t)$'s are identically distributed across t . In addition, the supports of $(X, \tilde{X}, Y)$ and $[X, \tilde{X}, Y]_{\text{swap}(\{j\})}$ are the same for each $j \in \{1, \dots, p\}$.

Condition 2. The knockoff generator $\kappa(\cdot, \cdot)$ is constructed independently of observed time series $\{\mathbf{x}_t, Y_t\}_{t=1}^n$.

Condition 1 is a basic regularity condition. Condition 2 may be relaxed if we use sample splitting to obtain an asymptotically independent training subsample for estimating the unknown covariate distribution and constructing the knockoff generator.

To ease the presentation, let $(Y, \mathbf{x}, \tilde{\mathbf{x}})$ be an independent copy of $(Y_1, \mathbf{x}_1, \tilde{\mathbf{x}}_1)$. As outlined in the last subsection, we allow certain deviation of $\kappa(\mathbf{x}, \cdot)$ from the one derived from the true covariate distribution of $\mathbf{x}$. Consequently, $\tilde{\mathbf{x}}$ generated from $\kappa(\mathbf{x}, \cdot)$ is only an *approximate* knockoff vector, which may violate the exchangeability condition in (2). The use of approximate knockoffs instead of ideal model-X knockoffs may incur a potential FDR inflation. Our Theorem 1, which imposes mild conditions on $\kappa(\cdot, \cdot)$, demonstrates that such FDR inflation can be controlled in time series applications. These results generalize the robust knockoffs inference results for iid observations

in Barber, Candès, and Samworth (2020) to the time series setting.

Let $\{\mathbf{x}_t^\pi, \tilde{\mathbf{x}}_t^\pi, Y_t^\pi\}_{t=1}^n$ be a sequence of iid random vectors such that $(\mathbf{x}_1^\pi, \tilde{\mathbf{x}}_1^\pi, Y_1^\pi)$ and $(\mathbf{x}_1, \tilde{\mathbf{x}}_1, Y_1)$ have the same distribution. Denote by $\mathcal{X}_k = \{\mathbf{x}_i, \tilde{\mathbf{x}}_i, Y_i\}_{i \in H_k}$ and $\mathcal{X}_k^\pi = \{\mathbf{x}_i^\pi, \tilde{\mathbf{x}}_i^\pi, Y_i^\pi\}_{i \in H_k}$ for each $k \in \{1, \dots, q+1\}$. Let $f_z(\cdot)$ be the density function of random vector $\mathbf{z}$.

Theorem 1. Let $\hat{S}$ be the set of variables selected by TSKI with Algorithm 1. Then under Conditions 1–2 and the assumption of positive T^{k*} 's in (5), we have

$$\text{FDR} \leq \inf_{\varepsilon > 0} \left[\tau^* e^\varepsilon + \sum_{k=1}^{q+1} \mathbb{P}(\max_{1 \leq j \leq p} \widehat{\text{KL}}_j^{k\pi} > \varepsilon) \right] + \sum_{k=1}^{q+1} \sup_{\mathcal{D} \in \mathcal{R}^{\#H_k \times (2p+1)}} |\mathbb{P}(\mathcal{X}_k \in \mathcal{D}) - \mathbb{P}(\mathcal{X}_k^\pi \in \mathcal{D})|, \quad (6)$$

where $0 < \tau^* < 1$ is the target FDR level and for each $1 \leq k \leq q+1$ and $1 \leq j \leq p$,

$$\widehat{\text{KL}}_j^{k\pi} = \sum_{i \in H_k} \log \left(\frac{f_{X_j, \tilde{X}_j, \mathbf{x}_{-j}, \tilde{\mathbf{x}}_{-j}}(X_{ij}^\pi, \tilde{X}_{ij}^\pi, \mathbf{x}_{-ij}^\pi, \tilde{\mathbf{x}}_{-ij}^\pi)}{f_{X_j, \tilde{X}_j, \mathbf{x}_{-j}, \tilde{\mathbf{x}}_{-j}}(\tilde{X}_{ij}^\pi, X_{ij}^\pi, \mathbf{x}_{-ij}^\pi, \tilde{\mathbf{x}}_{-ij}^\pi)} \right). \quad (7)$$

An example is provided in Section A.1 of the supplementary material, where the KL divergence term on the right-hand side of (6) asymptotically vanishes. The KL divergence term can be further simplified provided that Condition 3, adapted from Definition 1 in Barber, Candès, and Samworth (2020), is satisfied. This condition concerns the knockoff generator $\kappa(\cdot, \cdot)$ and p additional coordinate-wise knockoff generators $\kappa_j : \mathbb{R}^{p-1} \times \mathcal{R} \mapsto \mathbb{R}$ for $j \in \{1, \dots, p\}$, where each $\kappa_j(\mathbf{x}_{-j}, \cdot)$ approximates the conditional distribution of X_j given $\mathbf{x}_{-j}$.

Condition 3 (Definition 1 in Barber, Candès, and Samworth (2020)). For each $1 \leq j \leq p$, 1) if $\tilde{\mathbf{z}} = (\tilde{Z}_1, \dots, \tilde{Z}_p)^T$ is sampled from the conditional distribution $\kappa((X_1, \dots, X_{j-1}, \tilde{X}_j^\dagger, X_{j+1}, \dots, X_p), \cdot)$, then $(\tilde{X}_j^\dagger, \tilde{Z}_j, \mathbf{x}_{-j}, \tilde{\mathbf{z}}_{-j})$ and $(\tilde{Z}_j, \tilde{X}_j^\dagger, \mathbf{x}_{-j}, \tilde{\mathbf{z}}_{-j})$ have the same distribution, where $\tilde{X}_j^\dagger$ is sampled from $\kappa_j(\mathbf{x}_{-j}, \cdot)$; and 2) the density function of the distribution of $(\tilde{X}_j^\dagger, \tilde{Z}_j, \mathbf{x}_{-j}, \tilde{\mathbf{z}}_{-j})$ exists.

Corollary 1. Assume that all the conditions of Theorem 1 hold. If further Condition 3 is satisfied, then (6) holds with

$$\widehat{\text{KL}}_j^{k\pi} = \sum_{i \in H_k} \log \left(\frac{f_{X_j | \mathbf{x}_{-j}}(X_{ij}^\pi | \mathbf{x}_{-ij}^\pi) f_{\tilde{X}_j^\dagger | \mathbf{x}_{-j}}(\tilde{X}_{ij}^\pi | \mathbf{x}_{-ij}^\pi)}{f_{X_j | \mathbf{x}_{-j}}(\tilde{X}_{ij}^\pi | \mathbf{x}_{-ij}^\pi) f_{\tilde{X}_j^\dagger | \mathbf{x}_{-j}}(X_{ij}^\pi | \mathbf{x}_{-ij}^\pi)} \right),$$

where $\tilde{X}_j^\dagger$'s are given in Condition 3 and $f_{\mathbf{z}_1 | \mathbf{z}_2}(\mathbf{z}_1 | \mathbf{z}_2) = f_{\mathbf{z}_1, \mathbf{z}_2}(\mathbf{z}_1, \mathbf{z}_2) [f_{\mathbf{z}_2}(\mathbf{z}_2)]^{-1}$ is the conditional probability density function of $\mathbf{z}_1$ given $\mathbf{z}_2$.

It is seen that when Condition 3 is met, the FDR inflation can be measured by the Kullback–Leibler (KL) divergence between the conditional distributions of $\tilde{X}_j^\dagger | \mathbf{x}_{-j}$ and $X_j | \mathbf{x}_{-j}$, where the former is described by the coordinate-wise knockoff generator $\kappa_j(\mathbf{x}_{-j}, \cdot)$. When these two conditional distributions are identical, the knockoff generator $\kappa(\cdot, \cdot)$ satisfying Condition 3 reduces

to the ideal knockoff generator in model-X knockoffs Candès et al. (2018). In this sense, [Condition 3](#) relaxes the requirement of knowing the exact covariate distribution.

We borrow the Gaussian example discussed in Barber, Candès, and Samworth (2020) to help understand [Condition 3](#). Consider the Gaussian knockoff generator described around (8). Lemma 4 of Barber, Candès, and Samworth (2020) demonstrates that $\kappa(\cdot, \cdot)$ and $\kappa_j(\cdot, \cdot)$ satisfy [Condition 3](#) if the former is chosen as

$$\kappa(\mathbf{x}_t, \cdot) \stackrel{d}{\sim} N(\mathbf{x}_t - \text{diag}(\hat{\Sigma})\hat{\Theta}\mathbf{x}_t, 2\text{diag}(\hat{\Sigma}) - \text{diag}(\hat{\Sigma})\hat{\Theta}\text{diag}(\hat{\Sigma})), \quad (8)$$

and each $\kappa_j(\mathbf{x}_{-j}, \cdot)$ follows the distribution $N(\mathbf{x}_{-j}^T \hat{\Theta}_{-jj} \hat{\Theta}_{jj}^{-1}, \hat{\Theta}_{jj}^{-1})$, where $\hat{\Theta}$ is a positive definite estimate of $[\mathbb{E}(\mathbf{x}\mathbf{x}^T)]^{-1}$ constructed independently from the observed data $(\mathbf{y}, \mathbf{X})$, $\hat{\Theta}_{-jj}$ represents the j th column of $\hat{\Theta}$ with the j th component removed, and $\hat{\Theta}_{jj}$ is the j th diagonal entry of $\hat{\Theta}$. Here, for simplicity, we assume that $\mathbf{x}_t$ has mean zero. [Corollary 1](#) indicates that the asymptotic FDR control depends on the estimation accuracy of $\hat{\Theta}$. See Section A.4 for how we estimate the precision matrix in practice. Meanwhile, see Barber, Candès, and Samworth (2020) for other examples of knockoffs generators satisfying [Condition 3](#).

Under [Condition 4](#), a simple upper bound for the second term on the right-hand side of (6) can be derived; this result is formally summarized in [Corollary 2](#). An example satisfying [Condition 4](#) is provided in [Section 2.4](#).

Condition 4 (h-step β -mixing with exponential decay). The covariate process $\{\mathbf{x}_t\}$ is a p -dimensional stationary Markov chain with a transition kernel $p : \mathbb{R}^p \times \mathcal{R}^p \mapsto \mathbb{R}$ and a stationary distribution π . There exist a positive integer h , a measurable function $V : \mathbb{R}^p \rightarrow [0, \infty)$, and some constants $0 \leq \rho < 1$ and $0 < C_0 < \infty$ such that for each $\vec{x} \in \mathbb{R}^p$, $\|p^h(\vec{x}, \cdot) - \pi(\cdot)\|_{TV} \leq C\rho^h V(\vec{x})$, where $C > 0$ is some constant with $C_0 \geq C \int_{\mathbb{R}^p} V(\vec{x}) d\pi(\vec{x})$ and $\|\cdot\|_{TV}$ denotes the total variation (TV) norm associated with measures. Moreover, for each $\vec{x} \in \mathbb{R}^p$, $p(\vec{x}, \cdot)$ is absolutely continuous with respect to the Lebesgue measure.

Corollary 2. Assume that all the conditions of [Theorem 1](#) hold. If further $\{\mathbf{x}_t\}_{t \geq 1}$ satisfies [Condition 4](#) with q -step and constants $C_0 > 0$ and $0 \leq \rho < 1$, and Y_i is $\mathbf{x}_{t+1}$ -measurable, then (6) holds with

$$\begin{aligned} & \sum_{k=1}^{q+1} \sup_{\mathcal{D} \in \mathcal{R}^{\#H_k \times (2p+1)}} |\mathbb{P}(\mathcal{X}_k \in \mathcal{D}) - \mathbb{P}(\mathcal{X}_k^\pi \in \mathcal{D})| \\ & \leq C_0 \times \rho^q \times n. \end{aligned} \quad (9)$$

Moreover, when $(Y_t, \mathbf{x}_t)$'s are iid, (9) holds with $\rho = 0$.

As shown in [Corollary 2](#), in the iid data setting where $\rho = 0$, the FDR upper bound in (6) replicates the result in Barber, Candès, and Samworth (2020) when $q = 0$ (i.e., no subsampling). With serial dependence and general q , from (6) and (9), we observe an interesting tradeoff between the KL divergence and the TV upper bound as q changes. First, in view of the expression in [Corollary 1](#), $\mathbb{P}(\max_{1 \leq j \leq p} \widehat{\text{KL}}_j^{k\pi} > \varepsilon)$ depends on q in a complicated way via the subsample size $\lfloor n/(q+1) \rfloor$. In

addition, since the probability is upper bounded by 1, the term $\sum_{k=1}^{q+1} \mathbb{P}(\max_{1 \leq j \leq p} \widehat{\text{KL}}_j^{k\pi} > \varepsilon)$ increases at most linearly with q . On the other hand, [Corollary 2](#) suggests that the TV upper bound decreases exponentially with q . Despite this tradeoff, it is unreasonable to determine the optimal choice of q by directly analyzing these upper bounds since they can be conservative. We leave the theoretical investigation of the optimal q for future research. Our empirical study suggests that $q = 1$ or 2 often yields satisfactory finite-sample control of the FDR, as long as each subsample is reasonably large.

To provide a concrete time series example where our theory in this section applies, we analyze the Gaussian ARX model ([Example 3](#) in the next subsection) with $\Sigma = \mathbb{E}(\mathbf{x}\mathbf{x}^T)$ assumed to be known, and $q = \lceil (\log n)^{1+\delta} \rceil$ and $p = O(n^{K_0})$ for some constants $\delta > 0$ and $K_0 > 0$. We show in Section A.1 that the KL divergence is zero and prove in [Section 2.4](#) that [Condition 4](#) is satisfied. Thus, the FDR upper bound in [Theorem 1](#) becomes $\tau^* + C_0 n \rho^q$. When Σ is unknown and needs to be estimated, it is possible to obtain an upper bound using an estimated covariance matrix formed from a large independent learning sample using the proof idea for Lemma 5 of (Barber, Candès, and Samworth 2020).

When moving beyond the Gaussian time series covariates, for example, Model 1 in [Section 4](#), we assume the high-level condition $\sum_{k=1}^{q+1} \mathbb{P}(\max_{1 \leq j \leq p} \widehat{\text{KL}}_j^{k\pi} > \varepsilon) = o(1)$ and the β -mixing condition as in [Corollary 2](#). Under these assumptions, our theory applies. Despite these technical assumptions, our simulation results suggest that the FDR can be controlled in broad model settings with in-sample estimated covariate distribution. Theoretical justification of such empirical results for time series data is highly challenging and is left for future investigation.

2.4. Stationary Processes Satisfying Condition 4

We provide a linear time series example that satisfies [Condition 4](#). Additional examples, including nonlinear ones, are provided in Section A.2 of the supplementary material.

Example 3 (Autoregressive models with exogenous variables (ARX)). For each t , define

$$Y_t = \sum_{j=1}^{k_1} \alpha_j Y_{t-j} + \sum_{l=1}^{k_2} \sum_{j=1}^{k_{3l}} \beta_j^{(l)} H_{t-j+1}^{(l)} + \varepsilon_t$$

with $H_t^{(l)} = \varepsilon_t^{(l)} + \sum_{j=1}^{k_{3l}} \beta_j^{(l)} H_{t-j}^{(l)}$, where for some positive constants L_0, L_1 , and L_2 , it satisfies that $1 - \sum_{j=1}^{k_{3l}} \beta_j^{(l)} z^j \neq 0$ and $1 - \sum_{j=1}^{k_1} \alpha_j z^j \neq 0$ for each $|z| \leq 1 + L_0$ and each $l \in \{1, \dots, k_2\}$. Additionally, $\sum_{l=1}^{k_2} \sum_{j=1}^{k_{3l}} |\beta_j^{(l)}| < L_1$ and $k_1, k_{31}, \dots, k_{3k_2} < L_2$. Here, k_1, k_2 , and $k_{31}, \dots, k_{3k_2}$ are all positive integers. Moreover, $(\varepsilon_t^{(1)}, \dots, \varepsilon_t^{(k_2)}, \varepsilon_t)$'s are $(k_2 + 1)$ -dimensional iid Gaussian random vectors with zero mean and positive definite covariance matrix Σ_0 that satisfy $\mathbb{E}(\varepsilon_t \varepsilon_t^{(l)}) = 0$ for each l .

[Example 3](#) is a benchmark model for describing the behavior of macroeconomic variables. It is seen that the covariate vector

with respect to response Y_t is $\mathbf{x}_t = (Y_{t-1}, \dots, Y_{t-k_1}, \mathbf{h}_t^T)^T$ with $\mathbf{h}_t = (H_t^{(1)}, \dots, H_{t-k_{31}+1}^{(1)}, \dots, H_t^{(k_2)}, \dots, H_{t-k_{3k_2}+1}^{(k_2)})^T$. It has been shown (e.g., An and Huang (1996)) that the stationary process $\{\mathbf{x}_t\}$ in Example 3 with fixed dimensionality satisfies Condition 4 with h -step and some constants ρ, C_0 for each positive integer h . When the dimensionality of $\mathbf{x}_t$ increases, certain growth conditions such as (10) on the value of h and the dimensionality of $\mathbf{x}_t$ are needed for Condition 4 to hold. For this reason, we make the dependence of the stationary processes on h explicit: hereafter $\{\mathbf{x}_t^{(h)}\}$ denotes a p_h -dimensional stationary process satisfying Condition 4 for $h \geq 1$, as guaranteed by Proposition 1.

Proposition 1. Let $\{\mathbf{x}_t^{(h)}\}$ be a sequence of p_h -dimensional linear process in Example 3 with constant L_i 's and uniformly positive definite $\Sigma_0^{(h)}$'s. Assume that for some constant $C_2 > 0$ and sufficiently small $s_2 > 0$,

$$\sup_{h>0} \{p_h \exp(-s_2 h)\} \leq C_2. \quad (10)$$

Then for some constants $0 \leq \rho < 1$ and $C_0 > 0$ and all large h , $\{\mathbf{x}_t^{(h)}\}$ satisfies Condition 4.

In view of Theorem 1 and Proposition 1, if we assume Example 3 with $p = p_{q_n} = O(n^{K_0})$ and subsample with $q = q_n = \lceil (\log n)^{1+\delta} \rceil$, where $K_0 > 0$ and $\delta > 0$ are some constants, then the β -mixing convergence rate required by Theorem 1 is satisfied by the subsampled data for all large n .

3. Power Analysis Under Generalized Linear Time Series Models

Since the selection power of any procedure depends on the signal strength, we showcase the power analysis using the GLM time series model where the signal strength can be measured conveniently by the regression coefficients. Correspondingly, we consider the LCD knockoff statistic in Example 1 because Lasso is popularly used in high-dimensional GLM regression.

The canonical GLM has the conditional mean function

$$\mathbb{E}(Y_t | \mathbf{x}_t) = g(\mathbf{x}_t^T \vec{\beta}^o), \quad (11)$$

where $\vec{\beta}^o$ is a vector of unknown coefficients and $g(\cdot)$ is the derivative of a differentiable function $r(\cdot)$. The inverse function of $r'(\cdot)$, denoted as $g^{-1}(\cdot)$, is referred to as the canonical link function. Commonly used canonical link functions include: (1) the identity link $g^{-1}(\mu) = \mu$ for the linear model, (2) the logit link $g^{-1}(\mu) = \log(\mu/(1-\mu))$ with $0 < \mu < 1$ for the logistic model, and (3) the log link $g^{-1}(\mu) = \log \mu$ with $\mu > 0$ for the Poisson model.

To form the LCD knockoff statistics, we first fit the following GLM Lasso regression

$$(\hat{\beta}_1, \dots, \hat{\beta}_{2p})^T = \arg \min_{\vec{\beta} \in \mathbb{R}^{2p}} \left\{ n^{-1} \sum_{t=1}^n 2 \left(-Y_t \times (\mathbf{x}_t^T, \tilde{\mathbf{x}}_t^T) \vec{\beta} + r \left((\mathbf{x}_t^T, \tilde{\mathbf{x}}_t^T) \vec{\beta} \right) \right) + \lambda_n \sum_{j=1}^{2p} |\beta_j| \right\}, \quad (12)$$

where $\lambda_n \geq 0$ is the regularization parameter. Define $\vec{\beta}^* := (\beta_1^*, \dots, \beta_{2p}^*)^T$ with $(\beta_1^*, \dots, \beta_p^*)^T = \vec{\beta}^o$ and $\beta_{p+1}^* = \dots = \beta_{2p}^* = 0$. By the conditional independence $Y_t \perp\!\!\!\perp \tilde{\mathbf{x}}_t | \mathbf{x}_t$, we have $\mathbb{E}(Y_t | \mathbf{x}_t, \tilde{\mathbf{x}}_t) = \mathbb{E}(Y_t | \mathbf{x}_t) = g(\mathbf{x}_t^T \vec{\beta}^o) = g((\mathbf{x}_t^T, \tilde{\mathbf{x}}_t^T) \vec{\beta}^*)$. Thus, (12) estimates the population parameter vector $\vec{\beta}^*$. When the link function is the identity, the GLM Lasso estimate becomes the linear Lasso estimate.

Extensive research has been conducted to study the performance of Lasso with time series data. See, for example, Adamek, Smeekes, and Wilms (2023); Basu and Michailidis (2015); Han, Tsay, and Wu (2023); Kock and Callot (2015); Medeiros and Mendes (2016); Wong, Li, and Tewari (2020). Specifically, error bounds for the linear Lasso estimates in ARX models with conditionally heteroscedastic errors were considered in Adamek, Smeekes, and Wilms (2023); Medeiros and Mendes (2016); Wong, Li, and Tewari (2020). Moreover, Han, Tsay, and Wu (2023) established error bounds and support recovery guarantees of the GLM Lasso when both the response and covariate vector are stationary time series with dependence measures (as defined therein) satisfying certain summability conditions. Our power analysis needs the following technical conditions.

Condition 5. For some constant $c_0 > 0$ and sequence $k_{3n} > 0$ with $\lim_{n \rightarrow \infty} k_{3n} = 0$, it holds that $\mathbb{P} \left(\sum_{j=1}^{2p} |\hat{\beta}_j - \beta_j^*| \leq c_0 (\#S^*) \lambda_n \right) \geq 1 - k_{3n}$, where $S^* = \{j : |\beta_j^*| > 0\}$.

Condition 6. There exists some sequence $k_{1n} > 0$ with $k_{1n} q^{-1} \rightarrow \infty$ as $n \rightarrow \infty$ such that $\min_{j \in S^*} |\beta_j^*| > k_{1n} \lambda_n$.

Condition 7. For some constant $c_1 \in (0, 1)$ and sequence $\{k_{2n}\}$ with $\lim_{n \rightarrow \infty} k_{2n} = 0$, it holds that $2(\tau_1 \#S^*)^{-1} < c_1$ and $\mathbb{P}(\#\{j : W_j^k \geq T^k\} \geq c_1 (\#S^*)) \geq 1 - k_{2n}$ for $k \in \{1, \dots, q+1\}$.

We note that Condition 5 and (13) are the L_1 - and L_2 -estimation error bounds for the linear Lasso, which were established in previous studies (Adamek, Smeekes, and Wilms 2023; Basu and Michailidis 2015; Kock and Callot 2015; Medeiros and Mendes 2016; Wong, Li, and Tewari 2020) in time series settings under various different model assumptions with model-specific choice of λ_n . In these studies, k_{1n}, k_{2n} , and k_{3n} typically decrease at polynomial rates as n increases. Additionally, the GLM Lasso estimation error bounds were established in Han, Tsay, and Wu (2023). As a concrete example, for the model in Example 3 with dimensionality p increasing at most exponentially with sample size n , we can prove that Condition 5 and (13) hold with $\lambda_n = O([\log p]^{3/2} n^{-1/2+\alpha})$ for some $\alpha \in (0, 1/2)$ following the same proof as that for Theorem 3.1 in Han, Tsay, and Wu (2023) if, say, the Gaussian knockoff generator (3) is used. Since the proof is a straightforward extension, we opt to omit it to save space.

Condition 6 shares similarities with the often-imposed beta-min assumption $\min_{j \in S^*} |\beta_j^*| > \sqrt{\#S^*} \times \lambda_n$ for ensuring Lasso's model selection consistency. If q is chosen as $O((\log n)^{1+\delta})$ as suggested by Proposition 1 and $\sqrt{\#S^*} \gg (\log n)^{1+\delta}$, then Condition 6 becomes less restrictive than the beta-min condition. Condition 7 is a technical assumption that can be proved using

the same derivations as in Fan et al. (2020) under [Condition 5](#) and the following L_2 -estimation error bound condition

$$\mathbb{P} \left(\left[\sum_{j=1}^{2p} (\hat{\beta}_j - \beta_j^*)^2 \right]^{\frac{1}{2}} \leq c_0 \sqrt{\#S^* \lambda_n} \right) \geq 1 - k_{2n}. \quad (13)$$

For technical simplicity in power analysis, we assume that there are no ties in the magnitude of nonzero knockoff statistics and Lasso solutions. Furthermore, we assume that T^k 's in (5) satisfy $\max_{1 \leq k \leq q+1} \{T^k\} < \infty$ almost surely.

Theorem 2. Assume $S^* \subset \{1, \dots, p\}$ with $\#S^* > 0$. Let $\hat{S}$ be returned by [Algorithm 1](#) with $\tau_1, \tau_* \in (0, 1)$ and the LCD knockoff statistics (see [Example 1](#)). Assume that [Conditions 6–7](#) are satisfied and [Condition 5](#) holds for the Lasso estimates applied to each subsample H_k in [Algorithm 1](#). Then it holds that for all large n ,

$$\mathbb{P} \left(\hat{S} = \emptyset \cup \left\{ \frac{\#(S^* \cap \hat{S})}{\#S^*} \geq 1 - 4c_0(1+q)k_{1n}^{-1} \right\} \right) \geq 1 - (q+1) \times (k_{2n} + k_{3n}). \quad (14)$$

If we further assume that $\tau_1 = \tau^* \times K^{-1} \times (1 - 4(q+1)c_0k_{1n}^{-1})$ with some $K > 1$, then for all large n ,

$$\mathbb{E} \left(\frac{\#(S^* \cap \hat{S})}{\#S^*} \right) \geq \left(1 - \frac{(q+1)(\tau_1 + \theta_\varepsilon)K}{K-1} - (q+1) \times (k_{2n} + k_{3n}) \right) \times k_{4n}, \quad (15)$$

where $\{k_{4n}\}$ is some increasing sequence with $\lim_{n \rightarrow \infty} k_{4n} = 1$ and

$$\theta_\varepsilon = \inf \left\{ \theta \geq 0 : \max_{1 \leq k \leq q+1} \mathbb{E} \left(\frac{\#(\{j : W_j^k \geq T^k\} \cap (S^*)^c)}{\#\{j : W_j^k \geq T^k\} \vee 1} \right) \leq \tau_1 + \theta \right\}. \quad (16)$$

As previously discussed, k_{2n} and k_{3n} generally converge to zero at polynomial rates as n increases. Hence, setting $q = O((\log n)^{1+\delta})$ makes the right-hand side of (14) asymptotically approaching one. Thus, with asymptotic probability one, [Algorithm 1](#) either makes no discovery or has the percentage of correct discovery $\frac{\#(S^* \cap \hat{S})}{\#S^*}$ approaching one. The event of no discovery occurs when most individual knockoff filters from different subsamples select an abundantly large number of false positives so that the resulting e-values for all variables become too small to be selected. To further exclude the event $\{\hat{S} = \emptyset\}$, it is essential for $(\tau_1 + \theta_\varepsilon)q$ to be asymptotically vanishing (see (15)). To understand this, note that when $(\tau_1 + \theta_\varepsilon)q$ is sufficiently small, it effectively controls the false discovery proportion for each knockoff filter, consequently providing an upper bound on $\sum_{s=1}^p \mathbf{1}_{\{W_s^k \leq -T^k\}}$ for each $k = 1, \dots, q+1$. This further entails that the j th e-value statistic given in [Algorithm 1](#) is sufficiently large for each $j \in S^*$ so that it can be selected by the e-value procedure and hence $\hat{S} \neq \emptyset$. In practical implementation, in light of the definition of τ_1 , we can make $\tau_1 q$ asymptotically negligible by choosing $K \gg q = O((\log n)^{1+\delta})$.

4. Simulation Studies

We now investigate the finite-sample performance of the TSKI procedure with [Algorithm 1](#). Two selection methods considered in this section are TSKI with LCD in [Example 1](#) and TSKI with random forests MDA in [Example 2](#); we abbreviate them as TSKI-LCD and TSKI-MDA, respectively. We generate the approximate knockoff variables using the idea of the second-order approximation (Candès et al. 2018; Fan et al. 2020) via the knockoff generator (8); See Section A.4 for the estimation of $\hat{\Theta}$ and the choice of $\vec{s}$.

We compare with two benchmark methods. The first method selects features using the Benjamini–Yekutieli (BY) method (Benjamini and Yekutieli 2001). We choose not to use the Benjamini–Hochberg (BH) method (Benjamini and Hochberg 1995) because it assumes independence among the underlying p -values, which is not expected to hold true for time series applications. Due to the limited results on obtaining p -values for high-dimensional time series regression models, for the BY method, we calculated p -values by using the ordinary least squares method in our simulations when $n > p$. When $n < p$, the BY method is not applicable due to the lack of an effective p -value calculation method. The second benchmark method is the adaptive Lasso (Zou 2006; Medeiros and Mendes 2016), which was designed for feature selection without the aim of FDR control. These two approaches are abbreviated as LS-BY and adaLasso, respectively.

4.1. Simulation Settings

We consider the self-exciting threshold autoregressive model with exogenous variables (SETARX: Tong and Lim 1980), which is one of the most commonly used models for regime changes in time series data. Regime changes are frequently observed in economic time series (e.g., time series in [Figure 1](#)). It is well-known that SETARX with fixed dimensionality p satisfies the β -mixing [Condition 4](#) and is stationary; see, for example, An and Huang (1996). We present additional simulation experiments in Section A.4 of the supplementary material, where some other popular time series models such as the linear autoregressive model with exogenous variables and the autoregressive conditional heteroscedasticity model with exogenous variables (Engle 1982) are considered.

Model 1 (SETARX model). For each integer t and $\iota \in \{0, 5\}$, we define

$$Y_t = \begin{cases} \sum_{j=1}^2 (-0.5)^{j-1} \beta Y_{t-j} + 0.6(\sum_{j=1}^{\iota} H_{t,j} + \sum_{j=\iota+1}^{15} H_{t,j}) + \varepsilon_t, & \text{if } Y_{t-d} > r, \\ \sum_{j=1}^2 (-0.5)^{j-1} \beta Y_{t-j} + 0.6(-\sum_{j=1}^{\iota} H_{t,j} + \sum_{j=\iota+1}^{15} H_{t,j}) + \varepsilon_t, & \text{otherwise,} \end{cases}$$

where we choose $r = 0.7$ and $d = 1$ as the threshold value and the threshold lag, respectively.

We set the autoregressive coefficient $\beta = 0.7$, and choose the model errors ε_t 's as iid $N(0, 1)$. The time series covariates $H_{t,j}$'s are generated as $H_{t,j} = \eta \times H_{t-1,j} + \epsilon_{t,j}$ with $j \in \{1, \dots, 50\}$ and $\eta = 0.2$, where $(\epsilon_{t,1}, \dots, \epsilon_{t,50})$'s are iid Gaussian random vectors across t with zero mean and $\mathbb{E}(\epsilon_{t,k} \epsilon_{t,l}) = (0.2)^{|k-l|}$.

for all k, l . We formulate the covariate vector with respect to response Y_t as $\mathbf{x}_t = (Y_{t-1}, \dots, Y_{t-20}, \mathbf{h}_t, \mathbf{h}_{t-1}, \mathbf{h}_{t-2}, \mathbf{h}_{t-3}, \mathbf{h}_{t-4})$ with $\mathbf{h}_t = (H_{t,1}, \dots, H_{t,50})$, giving rise to $p = 270$. Throughout the simulation, we keep $p = 270$ while varying the sample size n across experiments with $n \in \{200, 300, 500, 1000\}$. Due to the space constraint, we move the results for $n = 300$ and 1000 to Section A.4.

It is seen that the mean function is comprised of two components: a piecewise linear function of the lags and the first ι time series covariates, along with a linear function involving $(H_{t,\iota+1}, \dots, H_{t,15})$ for $\iota \in \{0, 5\}$. The setup with $\iota = 5$ mimics a realistic and straightforward scenario where certain variables $H_{t,j}$, but not all, together with lagged variables Y_{t-j} , undergo changes in regime. The mean regression function of Model 1 is therefore nonlinear, as illustrated graphically in Figure 2. The relevant index set according to Definition 1 is $S_0 = \{1, 2, 21, \dots, 35\}$, while the null set $\mathcal{H}_0 = \{3, \dots, 20, 36, \dots, 270\}$.

For implementation, the target FDR level is set as $\tau^* = 0.2$, and the R packages `glmnet` and `randomForest` are used for calculating the Lasso estimates and the random forests MDA, respectively. The TSKI Algorithm 1 with parameters $q \in \{0, 1, 2\}$, $\tau^* = 0.2$, and $\tau_1 = \tau^*/(q+1)$ is used in our simulation. The empirical versions of the FDR and power are calculated as the sample averages of the false discovery proportion and true discovery proportion across 100 repetitions, respectively. The values of (p, q, n) are included in both Tables 1–2, and the R codes are available in the supplementary material.

4.2. Empirical Performance of TSKI

Table 1 presents the results for TSKI-LCD and TSKI-MDA with $q = 0$ and $q = 1$. The complete results, which also include $q = 2$

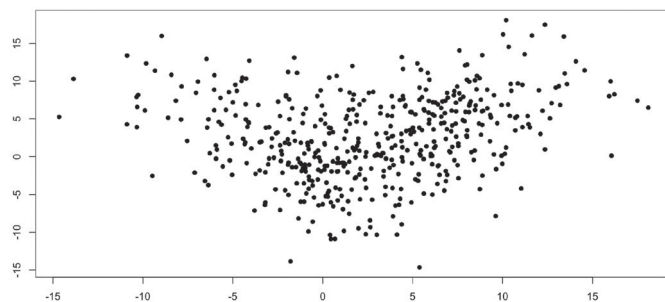


Figure 2. A graphical representation depicting Y_{t-1} on the x-axis and Y_t on the y-axis is provided under Model 1 with $(\eta, \iota) = (0.2, 5)$.

and a larger sample size $n = 1000$, are provided in Section A.4. It is seen from Table 1 that both methods control the FDR in finite samples below the target level of $\tau^* = 0.2$, while larger q gives lower selection power compared to that of $q = 0$ (i.e., no subsampling). The lower power for larger q is reasonable and caused by the small sample size in each subsample when fitting the time series regression model. The knockoffs method is empirically known to be conservative when the sample size is overly small. This conservative nature offsets the FDR inflation due to serial dependence in the small sample size scenario, which explains why FDR is still controlled even without subsampling. The extended Model 1 simulation presented in Section A.4 shows that for a larger sample size $n = 1000$, we start observing FDR inflation when $q = 0$. In addition, in the additional simulation examples presented in Section A.4, we observe severe FDR inflation when $q = 0$ in various scenarios (see Table 3, supplementary materials).

From Table 1, we see that for Model 1 with $\iota = 0$ (i.e., lower nonlinearity), the LCD-based method demonstrates superior performance over the MDA-based method in terms of power. Meanwhile, when $\iota = 5$ (i.e., higher nonlinearity), we observe from Table 1 that MDA outperforms LCD in power when the sample size is large (i.e., $n = 500$ and $q = 0$), an intuitive observation considering the nonparametric nature of the MDA measure. Indeed, the empirical performance of MDA decreases drastically in all settings as q increases, because of the smaller sample size when calculating the MDA measures.

The results from Table 1 highlight that TSKI-MDA may not be suitable for some real data applications, such as our study where only 60 observations are accessible for each inference window. They underline the need to explore ways to enhance selection power for nonlinear time series, especially with limited samples. On the other hand, Table 2 highlights that `adaLasso` demonstrates one of the highest selection powers among all four methods. However, `adaLasso` fails to control error rates under Model 1. This is expected because `adaLasso` is designed for the linear model and focuses on consistent model selection, lacking an error rate tuning parameter, such as the target FDR level. LS-BY is inapplicable to high-dimensional time series applications when $n < p$, such as our real data example in Section 5, due to the lack of a reliable p -value calculation method. It is important to emphasize that obtaining valid p -values under high-dimensional linear or nonlinear time series models presents a highly challenging and currently unresolved issue; such challenge is also reflected in the poor performance of LS-BY when p becomes comparable to n (i.e., $(n, p) = (300, 270)$).

Table 1. The simulation results on the empirical FDR and power for the TSKI with $\tau_1 = \tau^*/(q+1)$ and $\tau^* = 0.2$ under Model 1 in Section 4.1.

Method	$n/p/\eta/\iota$	q	FDR	Power	$n/p/\eta/\iota$	q	FDR	Power
TSKI-LCD	200/270/0.2/0	0	0.157	0.698	500/270/0.2/0	0	0.176	0.939
TSKI-LCD		1	0.026	0.051		1	0.099	0.872
TSKI-MDA		0	0.173	0.456		0	0.181	0.922
TSKI-MDA		1	0.026	0.028		1	0.092	0.550
TSKI-LCD	200/270/0.2/5	0	0.139	0.287	500/270/0.2/5	0	0.141	0.634
TSKI-LCD		1	0.023	0.019		1	0.086	0.267
TSKI-MDA		0	0.138	0.215		0	0.166	0.679
TSKI-MDA		1	0.012	0.011		1	0.084	0.216

NOTE: The results for $n \in \{300, 1000\}$ and $q = 2$ are given in Section A.4

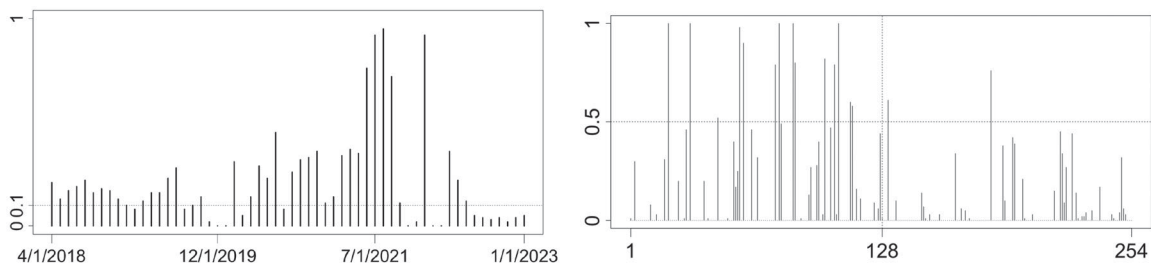


Figure 3. The left panel displays the averages of “having any selections” indicators over 100 repetitions, where the indicator at each rolling window is one if and only if any covariates are selected, and the x-axis indicates the ending time of each rolling window. The right panel shows the analogous results, but the indicator is one at each covariate index if that covariate is selected at any rolling window. The first 127 covariates are current time covariates, and the 128th to 254th covariates are one-month lag covariates in the AR(2) model. Covariates measuring similar economic values are clustered closer (see McCracken and Ng (2016) for detailed definitions of these covariates). The selection method here is the TSKI-LCD without subsampling ($q = 0$).

Table 2. Left panel: the simulation results on the empirical FDR and power for the adaptive Lasso (Zou 2006; Medeiros and Mendes 2016); there is no target FDR level for the adaptive Lasso. Right panel: the simulation results on the empirical FDR and power for the ordinary least squares + Benjamini–Yekutieli (BY (Benjamini and Yekutieli 2001)) with the target FDR level at 0.2; this approach does not apply to high-dimensional scenarios when $n < p$.

Adaptive Lasso			LS + BY		
$n/p/\eta/\iota$	FDR	Power	$n/p/\eta/\iota$	FDR	Power
200/270/0.2/0	0.520	0.964	200/270/0.2/0	–	–
300/270/0.2/0	0.468	0.997	300/270/0.2/0	0.000	0.001
500/270/0.2/0	0.657	1.000	500/270/0.2/0	0.027	0.763
200/270/0.2/5	0.604	0.705	200/270/0.2/5	–	–
300/270/0.2/5	0.563	0.786	300/270/0.2/5	0.018	0.006
500/270/0.2/5	0.677	0.891	500/270/0.2/5	0.026	0.276

It is worth pointing out that our additional simulation experiments in Section A.4 of the supplementary material reveal that: (a) TSKI-LCD and TSKI-MDA with $q = 1$ consistently control FDR below the target level, whereas setting $q = 0$ yields results occasionally exceeding the target level; and (b) the LS-BY method can have FDR exceeding the target level when data is simulated from autoregressive conditional heteroscedasticity models with exogenous variables and higher value of η (as in Model 1). See the supplementary material for full details.

Overall, our simulation experiments show that the time series FDR is controlled stably by the TSKI procedure with subsampling parameter $q = 1$ in finite samples, whereas the selection power depends on the sample size, the subsampling parameter $q \geq 0$, and the choice of the feature importance measure. Importantly, our theoretical results (Theorem 1 and Corollary 2) and simulation results show that TSKI is among the first approaches with theoretically justified FDR control for dependent data under flexible β -mixing condition in Condition 4. For implementation, we suggest that practitioners working on time series inference with limited sample sizes initiate their diagnosis by using TSKI-LCD with q set to 1, and generate knockoff variables using knockoff generator (8).

5. Real Data Application

We analyze the U.S. inflation series data described in the Introduction. The monthly economic time series, including numerous types of consumer price indices, unemployment

rates, and housing prices, can be obtained from the FRED-MD database (McCracken and Ng 2016)³ and the U.S. Bureau of Labor Statistics. These time series have been pre-processed following the instructions of the FRED-MD database to make them more stationary (McCracken and Ng 2016). To address the concern on potential nonstationarity over the entire time span, we break it into 58 rolling windows, each covering a five-year period. Motivated by our simulation study, we apply TSKI-LCD with $q \in \{0, 1\}$ described in Section 4.1 (with knockoff generator (8)) to each five-year rolling window to investigate the temporal relations between the inflation and other time series variables. The inflation at time t is defined as the adjusted consumer price index for all goods: $\text{Inflation}_t := \left(\frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100 \right) \%$, where CPI_t is the consumer price index for all goods at time (month) t . Each time series has a FRED-MD code. For example, CPIAUCSL is the FRED-MD code of the inflation series. To reduce randomness resulting from the use of the knockoffs construction, we repeat the inference procedure 100 times, and report the average results in Figure 3 with $q = 0$, where the left panel displays whether any significant covariates have been found at each rolling window period, and the right panel shows the selection frequency over 58 rolling window periods of the 127 time series covariates and their one month lags. That is, we model the one month ahead inflation series Inflation_{t+1} (response) using an AR(2) model with 127 time series covariates at current time t and 127 one month lags of these time series covariates. Consequently, the total covariate dimensionality is $p = 254$.

As can be seen in Figure 3, the TSKI-LCD with $q = 0$ identifies some active windows around the year 2021 (COVID) and 2022 (Russia-Ukraine conflict), with the selection frequency concentrating on a sparse set of covariates. The majority of the selected variables are covariates at the current time t . The top 10 most frequently selected covariates by the TSKI-LCD with $q = 0$ are employment-related series (HWI, CLAIMSx), consumption-related indices (ACOGNO, CPIAUCSL), housing-related series (PERMITS), U.S. bond yields (GS5, GS10), stock market indices (S.P.div.yield, S.P.500), and exchange rates (EXCAUSx) at their current time t , with their FRED-MD codes given in the parentheses.

³The website URL: <https://research.stlouisfed.org/econ/mccracken/fred-databases/>.

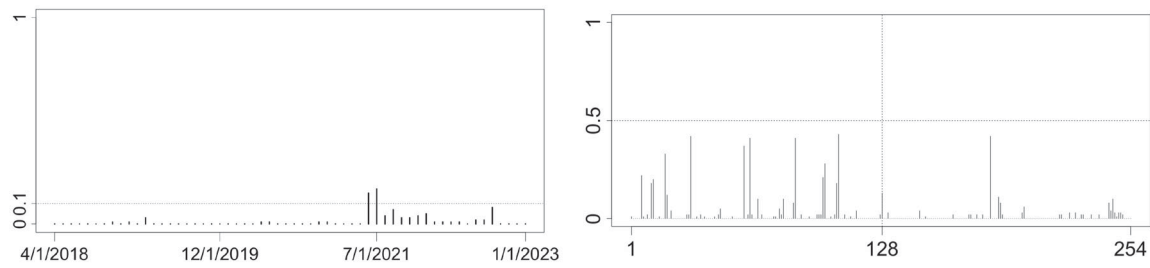
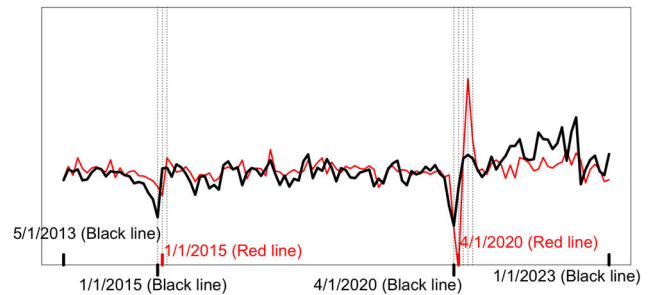


Figure 4. These two panels are analogous to those in Figure 3 but with $q = 1$ for the TSKI-LCD procedure.

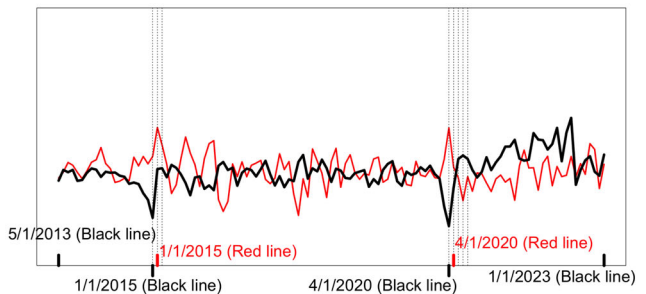
The simulation results in Section 4 suggest that the choice of $q = 1$ has better FDR control especially when the sample size is limited. Motivated by our simulation results, we next apply the TSKI-LCD with subsampling $q = 1$ in Figure 4 with the expectation of better FDR control. The results of Figure 4 are more conservative in comparison to those in Figure 3. Despite being conservative, these new results also suggest some active windows around the same periods as $q = 0$, and the selected variables are also mostly covariates at the current time t . In addition, most covariates selected by the TSKI-LCD with $q = 1$ belong to the set selected by the TSKI-LCD with $q = 0$. In particular, the top 10 selected covariates are CPIAUCSL, CLAIMS_x, PERMITS, AMDMUO_x, GS10, ACOGNO, EXCAUS_x, EXUSUK_x, INDPRO, and IPFUELS, where only PERMITS is one-month lag covariate at $t - 1$ in the AR(2) model. Among them, EXUSUK_x, AMDMUO_x, INDPRO, and IPFUELS are new in comparison to the list of the selected set when $q = 0$, where the first two are the U.S./U.K. exchange rate and another type of consumption-related index (the number of unfilled orders for durable goods), respectively, and the last two are industrial production indices that are related to consumption price indices. The difference in the selected sets of covariates is attributed to the fact that some economic covariates are designed to track similar economic factors and tend to be highly correlated.

The recent literature (Stock and Watson 2007) suggests that inflation forecasting is a difficult task in the sense that AR models with additional time series covariates rarely outperform simple AR models with only inflation lags. In other words, conditional on the lagged inflation series, additional covariates do not carry strong signals in inflation forecasting. The fact that the TSKI-LCD with $q = 1$ selects only a few time series covariates indeed supports such an argument. It is also interesting to notice that the stock market indices are not considered as important covariates by the TSKI-LCD with subsampling (i.e., $q = 1$), but are selected as important covariates when $q = 0$. In particular, the selection frequencies of S&P dividend yields and S&P 500 (both at lag $t - 1$) are 100% and 79%, respectively, in Figure 3, while only 5% and 1%, respectively, in Figure 4, suggesting that stock market indices could be spurious findings.

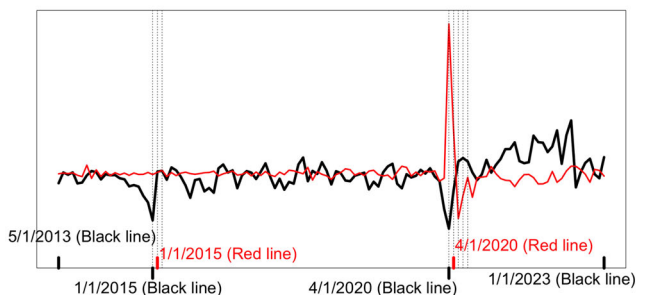
The selection results by the TSKI-LCD motivate us to further investigate the dependency of inflation on a few time series covariates. In Figure 5, we plot three selected series, namely ACOGNO, EXCAUS_x, and CLAIMS_x, which are among the top 10 lists both when $q = 0$ and $q = 1$. These three selected covariates are all at their current time t in the AR(2) model. ACOGNO is the number of new orders for consumer goods, which is an important consumption index; EXCAUS_x is the exchange rate from the U.S. dollar to the Canadian dollar, and



(a) The red curve is ACOGNO



(b) The red curve is EXCAUS_x



(c) The red curve is CLAIMS_x

Figure 5. The black curves in the three panels are the inflation series at time $t + 1$. The red curve in panel (a) is the number of new orders for consumer goods at time t , the red curve in panel (b) indicates the U.S./Canada exchange rate at time t , and the red curve in panel (c) is the U.S. initial claims for unemployment benefits at time t . All curves here are standardized and adjusted for visual comparison, and hence the values of these time series are not reported on the y-axis.

CLAIMS_x is the initial claim for unemployment benefits. We also include the inflation series at time $t + 1$ in the same period (the black curve in each panel). For better visual comparison, all curves in Figure 5 have been standardized and adjusted.

A visual inspection of Figure 5 shows that the impacts of the COVID-19 pandemic in April 2020 on the U.S. economy are stronger than those of the gasoline price shock in January 2015. This potentially explains why our empirical findings of significant covariates concentrate mostly on this period. From Figure 5, we see that the gasoline price shock in January 2015 affects the consumption index series ACOGNO more mildly compared to the impact of COVID-19 in April 2020. Also, although there is some variation in the exchange rate EXCAUSx after January 2015, it is unclear whether such variation was caused by the gasoline price shock. In contrast, most of the U.S. economy's time series clearly responded to the pandemic to an unignorable degree. Particularly, the exchange rate and the number of initial claims dropped in March 2020, suggesting that these covariates were leading indicators of the inflation drop in April 2020. In summary, we have applied the newly suggested tool of TSKI to study the U.S. economy. Our empirical results illustrate the potential of the TSKI to obtain more instructive findings in real data applications.

Supplementary Materials

In the supplement, we provide the appendix of Section 2, additional simulation examples, the proofs of all main results and technical lemmas, and some additional technical details.

Disclosure Statement

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